# Maclaurin Series & Convergence of Series

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**ANSWER KEY — COMPLETE WORKED SOLUTIONS**

## Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

**1** [Foundational] • Target: The Maclaurin Series Formula

Find the coefficient of $x^2$ in the Maclaurin series of $e^{3x}$.

Final Answer: $\frac{9}{2}$. $e^{3x} = \sum \frac{(3x)^k}{k!} \Rightarrow \text{coeff of } x^2 = \frac{3^2}{2!} = \frac{9}{2}$

**2** [Foundational] • Target: Manipulating Standard Maclaurin Expansions

Find the coefficient of $x^3$ in the Maclaurin series of $e^{3x}$.

Final Answer: $\frac{9}{2}$. $e^{3x} = \sum \frac{(3x)^k}{k!} \Rightarrow \text{coeff of } x^3 = \frac{3^3}{3!} = \frac{9}{2}$

**3** [Foundational] • Target: The Ratio Test for Convergence

Find the coefficient of $x^3$ in the Maclaurin series of $\sin(2x)$.

Final Answer: $-\frac{4}{3}$. $\sin(2x) = 2x - \frac{(2x)^3}{3!} + \cdots \Rightarrow \text{coeff of } x^3 = -\frac{4}{3}$

**4** [Foundational] • Target: Radius and Interval of Convergence

Find the coefficient of $x^4$ in the Maclaurin series of $\cos(2x)$.

Final Answer: $\frac{2}{3}$. $\cos(2x) = 1 - \frac{(2x)^2}{2!} + \cdots \Rightarrow \text{coeff of } x^4 = \frac{2}{3}$

**5** [Foundational] • Target: The Maclaurin Series Formula

Find the coefficient of $x^3$ in the Maclaurin series of $\sin(x)$.

Final Answer: $-\frac{1}{6}$. $\sin(x) = x - \frac{(x)^3}{3!} + \cdots \Rightarrow \text{coeff of } x^3 = -\frac{1}{6}$

**6** [Foundational] • Target: Manipulating Standard Maclaurin Expansions

Find the coefficient of $x^4$ in the Maclaurin series of $\cos(x)$.

Final Answer: $\frac{1}{24}$. $\cos(x) = 1 - \frac{(x)^2}{2!} + \cdots \Rightarrow \text{coeff of } x^4 = \frac{1}{24}$

**7** [Foundational] • Target: The Ratio Test for Convergence

Find the coefficient of $x^2$ in the Maclaurin series of $\cos(x)$.

Final Answer: $-\frac{1}{2}$. $\cos(x) = 1 - \frac{(x)^2}{2!} + \cdots \Rightarrow \text{coeff of } x^2 = -\frac{1}{2}$

## Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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**8** [Intermediate] • Target: Radius and Interval of Convergence

Find the coefficient of $x^2$ in the Maclaurin series of $\sin(3x)$.

Final Answer: $0$. $\sin(3x) = 3x - \frac{(3x)^3}{3!} + \cdots \Rightarrow$ coeff of $x^2 = 0$

**9** [Intermediate] • Target: The Maclaurin Series Formula

Find the coefficient of $x^2$ in the Maclaurin series of $\sin(2x)$.

Final Answer: $0$. $\sin(2x) = 2x - \frac{(2x)^3}{3!} + \cdots \Rightarrow$ coeff of $x^2 = 0$

**10** [Intermediate] • Target: Manipulating Standard Maclaurin Expansions

Find the coefficient of $x^4$ in the Maclaurin series of $\sin(x)$.

Final Answer: $0$. $\sin(x) = x - \frac{(x)^3}{3!} + \cdots \Rightarrow$ coeff of $x^4 = 0$

**11** [Intermediate] • Target: The Ratio Test for Convergence

Find the coefficient of $x^2$ in the Maclaurin series of $\sin(x)$.

Final Answer: $0$. $\sin(x) = x - \frac{(x)^3}{3!} + \cdots \Rightarrow$ coeff of $x^2 = 0$

**12** [Intermediate] • Target: Radius and Interval of Convergence

Fourier series period $2\pi$: $a_0 =$ .

Final Answer: $\frac{1}{2\pi}\int_{-\pi}^{\pi} f(x),dx$. Applying AA HL — Additional Topics principles, a■=(1/2π)∫f(x)dx. Therefore, the verified correct option is $\frac{1}{2\pi}\int_{-\pi}^{\pi} f(x),dx$.

**13** [Intermediate] • Target: The Maclaurin Series Formula

A sphere's radius increases at $\frac{dr}{dt} = 3$ cm/s. Find $\frac{dV}{dt}$ (in terms of $\pi$) when $r = 5$ cm.

Final Answer: $300\pi$ cm$^3$/s. Working through this calculus step:
$V = \frac{4}{3}\pi r^3 \Rightarrow \frac{dV}{dt} = 4\pi r^2 \frac{dr}{dt} = 4\pi(5)^2(3) = 300\pi$

**14** [Intermediate] • Target: Manipulating Standard Maclaurin Expansions

A sphere's radius increases at $\frac{dr}{dt} = 4$ cm/s. Find $\frac{dV}{dt}$ (in terms of $\pi$) when $r = 6$ cm.

Final Answer: $576\pi$ cm$^3$/s. Working through this calculus step:
$V = \frac{4}{3}\pi r^3 \Rightarrow \frac{dV}{dt} = 4\pi r^2 \frac{dr}{dt} = 4\pi(6)^2(4) = 576\pi$

## Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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**15** [Challenging] • Target: The Ratio Test for Convergence • Cross-Lesson: 4.02 The Poisson Distribution

A sphere's radius increases at $\frac{dr}{dt} = 1$ cm/s. Find $\frac{dV}{dt}$ (in terms of $\pi$) when $r = 5$ cm.

Final Answer: $100\pi$ cm$^3$/s. Working through this calculus step:

$V = \frac{4}{3}\pi r^3 \Rightarrow \frac{dV}{dt} = 4\pi r^2 \frac{dr}{dt} = 4\pi(5)^2(1) = 100\pi$

**16** [Challenging] • Target: Radius and Interval of Convergence • Cross-Lesson: 5.01 Further Integration Techniques

A sphere's radius increases at $\frac{dr}{dt} = 1$ cm/s. Find $\frac{dV}{dt}$ (in terms of $\pi$) when $r = 8$ cm.

Final Answer: $256\pi$ cm$^3$/s. Working through this calculus step:

$V = \frac{4}{3}\pi r^3 \Rightarrow \frac{dV}{dt} = 4\pi r^2 \frac{dr}{dt} = 4\pi(8)^2(1) = 256\pi$

**17** [Challenging] • Target: The Maclaurin Series Formula • Cross-Lesson: 5.02 Differential Equations

A sphere's radius increases at $\frac{dr}{dt} = 4$ cm/s. Find $\frac{dV}{dt}$ (in terms of $\pi$) when $r = 4$ cm.

Final Answer: $256\pi$ cm$^3$/s. Working through this calculus step:

$V = \frac{4}{3}\pi r^3 \Rightarrow \frac{dV}{dt} = 4\pi r^2 \frac{dr}{dt} = 4\pi(4)^2(4) = 256\pi$

**18** [Challenging] • Target: Manipulating Standard Maclaurin Expansions • Cross-Lesson: 5.04 Graphs of Inverse Trigonometric Functions & Their Derivatives (AA HL)

A sphere's radius increases at $\frac{dr}{dt} = 2$ cm/s. Find $\frac{dV}{dt}$ (in terms of $\pi$) when $r = 4$ cm.

Final Answer: $128\pi$ cm$^3$/s. Working through this calculus step:

$V = \frac{4}{3}\pi r^3 \Rightarrow \frac{dV}{dt} = 4\pi r^2 \frac{dr}{dt} = 4\pi(4)^2(2) = 128\pi$

**19** [Challenging] • Target: The Ratio Test for Convergence • Cross-Lesson: 6.01 Matrices (AI HL)

A sphere's radius increases at $\frac{dr}{dt} = 2$ cm/s. Find $\frac{dV}{dt}$ (in terms of $\pi$) when $r = 7$ cm.

Final Answer: $392\pi$ cm$^3$/s. Working through this calculus step:

$V = \frac{4}{3}\pi r^3 \Rightarrow \frac{dV}{dt} = 4\pi r^2 \frac{dr}{dt} = 4\pi(7)^2(2) = 392\pi$

**20** [Challenging] • Target: Radius and Interval of Convergence • Cross-Lesson: 6.02 Graph Theory & Networks (AI HL)

A sphere's radius increases at $\frac{dr}{dt} = 3$ cm/s. Find $\frac{dV}{dt}$ (in terms of $\pi$) when $r = 7$ cm.

Final Answer: $588\pi$ cm$^3$/s. Working through this calculus step:

$V = \frac{4}{3}\pi r^3 \Rightarrow \frac{dV}{dt} = 4\pi r^2 \frac{dr}{dt} = 4\pi(7)^2(3) = 588\pi$

## Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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**21** [Foundational] • Target: The Maclaurin Series Formula

What is the general formula for the Maclaurin series expansion of a function $f(x)$?

**a.** $f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!}x^n = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \ldots$ ✓

**b.** $f(x) = \sum_{n=0}^{\infty} f^{(n)}(0)x^n$

**c.** $f(x) = \sum_{n=1}^{\infty} \frac{f(n)}{n!}x^n$

**d.** $f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(x)}{n!}x^n$

A Maclaurin series is a Taylor series centered at $x = 0$: $f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!}x^n$.

**22** [Foundational] • Target: The Maclaurin Series Formula

Find the first three non-zero terms of the Maclaurin expansion of $f(x) = \cos(2x)$.

**a.** $1 - 2x^2 + \frac{2}{3}x^4$ ✓

**b.** $1 - \frac{1}{2}x^2 + \frac{1}{24}x^4$

**c.** $1 - 4x^2 + 16x^4$

**d.** $2x - \frac{4}{3}x^3 + \frac{4}{15}x^5$

Standard series $\cos u = 1 - \frac{u^2}{2} + \frac{u^4}{24} - \ldots$. Substitute $u = 2x$:
$1 - \frac{(2x)^2}{2} + \frac{(2x)^4}{24} = 1 - 2x^2 + \frac{16x^4}{24} = 1 - 2x^2 + \frac{2}{3}x^4$.

**23** [Foundational] • Target: The Maclaurin Series Formula

What is the coefficient of $x^3$ in the Maclaurin series of $e^{-3x}$?

**a.** $-\frac{9}{2}$ ✓

**b.** $\frac{9}{2}$

**c.** $-9$

**d.** $-27$

$e^u = \sum \frac{u^n}{n!}$. For $n = 3$, $\frac{(-3x)^3}{3!} = \frac{-27x^3}{6} = -\frac{9}{2}x^3$. The coefficient is $-\frac{9}{2}$.

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**24** [Foundational] • Target: Manipulating Standard Maclaurin Expansions

Find the Maclaurin series for $\ln(1+3x)$ up to the $x^3$ term.

**a.** $3x - \frac{9}{2}x^2 + 9x^3$ ✓

**b.** $3x + \frac{9}{2}x^2 + 9x^3$

**c.** $x - \frac{1}{2}x^2 + \frac{1}{3}x^3$

**d.** $3x - 9x^2 + 27x^3$

Standard series $\ln(1+u) = u - \frac{u^2}{2} + \frac{u^3}{3} - \ldots$. With $u = 3x$: $(3x) - \frac{(3x)^2}{2} + \frac{(3x)^3}{3} = 3x - \frac{9}{2}x^2 + 9x^3$.

**25** [Foundational] • Target: Manipulating Standard Maclaurin Expansions

Evaluate $\lim_{x \to 0} \frac{e^x - 1 - x}{x^2}$ using Maclaurin series.

**a.** $\frac{1}{2}$ ✓

**b.** $1$

**c.** $0$

**d.** $\infty$

$e^x = 1 + x + \frac{x^2}{2} + \ldots$. Then $e^x - 1 - x = \frac{x^2}{2} + O(x^3)$. Dividing by $x^2$: $\frac{1}{2} + O(x) \to \frac{1}{2}$ as $x \to 0$.

**26** [Foundational] • Target: Manipulating Standard Maclaurin Expansions

Find the first two non-zero terms of the series for $\frac{1}{1+x^2}$.

**a.** $1 - x^2$ ✓

**b.** $1 + x^2$

**c.** $1 - x$

**d.** $x - x^3$

Geometric series $\frac{1}{1-u} = 1 + u + u^2 + \ldots$. With $u = -x^2$: $1 - x^2 + x^4 - \ldots$.

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**27** [Foundational] • Target: The Ratio Test for Convergence

What does the Ratio Test state for a positive series $\sum a_n$ when $L = \lim_{n\to\infty} \frac{a_{n+1}}{a_n}$?

**a. Converges if $L < 1$, diverges if $L > 1$, test is inconclusive if $L = 1$ ✓**

**b.** Converges if $L = 1$, diverges otherwise

**c.** Diverges if $L < 1$, converges if $L > 1$

**d.** Always inconclusive for power series

The Ratio Test establishes absolute convergence when the limiting ratio $L < 1$, divergence when $L > 1$, and provides no information when $L = 1$.

**28** [Foundational] • Target: The Ratio Test for Convergence

Determine the convergence of $\sum_{n=1}^{\infty} \frac{n}{3^n}$ using the Ratio Test.

**a. Converges, because $L = \lim \frac{n+1}{3^{n+1}} \cdot \frac{3^n}{n} = \frac{1}{3} < 1$ ✓**

**b.** Diverges, because $L = 3 > 1$

**c.** Inconclusive, because $L = 1$

**d.** Diverges, because the terms do not approach 0

$\lim_{n\to\infty} \frac{n+1}{n} \cdot \frac{3^n}{3^{n+1}} = 1 \cdot \frac{1}{3} = \frac{1}{3}$. Since $L = 1/3 < 1$, the series converges.

## Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

**29** [Intermediate] • Target: The Ratio Test for Convergence

What is the limit $L$ in the Ratio Test for the harmonic series $\sum_{n=1}^{\infty} \frac{1}{n}$?

**a. $1$ (test is inconclusive) ✓**

**b.** 0

**c.** ∞

**d.** 1/2

$\lim_{n\to\infty} \frac{1/(n+1)}{1/n} = \lim \frac{n}{n+1} = 1$. The Ratio Test is inconclusive ($L = 1$); the harmonic series diverges by the p-series test ($p = 1$).

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**30** [Intermediate] • Target: Radius and Interval of Convergence

Find the radius of convergence $R$ for the power series $\sum_{n=1}^{\infty} \frac{(2x)^n}{n}$.

**a.** $R = \frac{1}{2}$ ✓

**b.** $R = 2$

**c.** $R = 1$

**d.** $R = \infty$

Ratio test: $\left|\frac{(2x)^{n+1}}{n+1} \cdot \frac{n}{(2x)^n}\right| = 2|x|\frac{n}{n+1} \to 2|x|$. For convergence: $2|x| < 1 \Rightarrow |x| < 1/2$. Thus $R = 1/2$.

**31** [Intermediate] • Target: Radius and Interval of Convergence

What is the radius of convergence of the Maclaurin series for $e^x$?

**a.** $R = \infty$ ✓

**b.** $R = 1$

**c.** $R = e$

**d.** $R = 0$

The exponential series converges for all complex numbers $x \in \mathbb{C}$; its radius of convergence is infinite ($R = \infty$).

**32** [Intermediate] • Target: Radius and Interval of Convergence

For $\sum_{n=0}^{\infty} x^n$, what is the exact interval of convergence?

**a.** $(-1, 1)$ ✓

**b.** $[-1, 1]$

**c.** $(-1, 1]$

**d.** $[-1, 1)$

Geometric series with ratio $x$: converges if and only if $|x| < 1$. At endpoints $x = \pm 1$, terms do not approach 0, so neither endpoint converges.

**33** [Intermediate] • Target: The Maclaurin Series Formula

A sphere's radius increases at $\frac{dr}{dt} = 1$ cm/s. Find $\frac{dV}{dt}$ (in terms of $\pi$) when $r = 4$ cm.

**a.** $24\pi$ cm$^3$/s

**b.** $74\pi$ cm$^3$/s

**c.** $64\pi$ cm$^3$/s ✓

**d.** $44\pi$ cm$^3$/s

Working through this calculus step: $V = \frac{4}{3}\pi r^3 \Rightarrow \frac{dV}{dt} = 4\pi r^2 \frac{dr}{dt} = 4\pi(4)^2(1) = 64\pi$

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**34** [Intermediate] • Target: Manipulating Standard Maclaurin Expansions

In the expansion of $(1+x)^{10}$, find the ratio of the coefficient of $x^5$ to the coefficient of $x^4$.

**a.** $\frac{8}{5}$

**b.** $1$

**c.** $\frac{7}{5}$

**d.** $\frac{6}{5}$ ✓

Working through this complex-numbers step: $\frac{\binom{10}{5}}{\binom{10}{4}} = \frac{10-4}{4+1} = \frac{6}{5}$

**35** [Intermediate] • Target: The Ratio Test for Convergence

In the expansion of $(1+x)^{10}$, find the ratio of the coefficient of $x^7$ to the coefficient of $x^6$.

**a.** $\frac{4}{7}$ ✓

**b.** $\frac{3}{7}$

**c.** $\frac{5}{7}$

**d.** $\frac{6}{7}$

Working through this complex-numbers step: $\frac{\binom{10}{7}}{\binom{10}{6}} = \frac{10-6}{6+1} = \frac{4}{7}$

**36** [Intermediate] • Target: Radius and Interval of Convergence

In the expansion of $(1+x)^{10}$, find the ratio of the coefficient of $x^3$ to the coefficient of $x^2$.

**a.** $\frac{8}{3}$ ✓

**b.** $\frac{10}{3}$

**c.** $3$

**d.** $\frac{7}{3}$

Working through this complex-numbers step: $\frac{\binom{10}{3}}{\binom{10}{2}} = \frac{10-2}{2+1} = \frac{8}{3}$

**37** [Intermediate] • Target: The Maclaurin Series Formula

In the expansion of $(1+x)^{8}$, find the ratio of the coefficient of $x^6$ to the coefficient of $x^5$.

**a.** $\frac{1}{2}$ ✓

**b.** $3$

**c.** $\frac{3}{2}$

**d.** $1$

Working through this complex-numbers step: $\frac{\binom{8}{6}}{\binom{8}{5}} = \frac{8-5}{5+1} = \frac{1}{2}$

## Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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**38** [Challenging] • Target: Manipulating Standard Maclaurin Expansions • Cross-Lesson: 5.04 Graphs of Inverse Trigonometric Functions & Their Derivatives (AA HL)

In the expansion of $(1+x)^7$, find the ratio of the coefficient of $x^6$ to the coefficient of $x^5$.

**a.** 1

**b.** $\frac{1}{3}$ ✓

**c.** $\frac{2}{3}$

**d.** 2

Working through this complex-numbers step: $\frac{\binom{7}{6}}{\binom{7}{5}} = \frac{7-5}{5+1} = \frac{1}{3}$

**39** [Challenging] • Target: The Ratio Test for Convergence • Cross-Lesson: 6.01 Matrices (AI HL)

In the expansion of $(1+x)^6$, find the ratio of the coefficient of $x^4$ to the coefficient of $x^3$.

**a.** $\frac{5}{4}$

**b.** $\frac{1}{2}$

**c.** 1

**d.** $\frac{3}{4}$ ✓

Working through this complex-numbers step: $\frac{\binom{6}{4}}{\binom{6}{3}} = \frac{6-3}{3+1} = \frac{3}{4}$

**40** [Challenging] • Target: Radius and Interval of Convergence • Cross-Lesson: 6.02 Graph Theory & Networks (AI HL)

In the expansion of $(1+x)^8$, find the ratio of the coefficient of $x^5$ to the coefficient of $x^4$.

**a.** $\frac{6}{5}$

**b.** $\frac{4}{5}$ ✓

**c.** 1

**d.** $\frac{3}{5}$

Working through this complex-numbers step: $\frac{\binom{8}{5}}{\binom{8}{4}} = \frac{8-4}{4+1} = \frac{4}{5}$

**41** [Challenging] • Target: The Maclaurin Series Formula • Cross-Lesson: 6.03 Markov Chains (AI HL)

In the expansion of $(1+x)^9$, find the ratio of the coefficient of $x^8$ to the coefficient of $x^7$.

**a.** $\frac{3}{4}$

**b.** $\frac{1}{4}$ ✓

**c.** $\frac{1}{2}$

**d.** $\frac{3}{2}$

Working through this complex-numbers step: $\frac{\binom{9}{8}}{\binom{9}{7}} = \frac{9-7}{7+1} = \frac{1}{4}$

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**42** [Challenging] • Target: Manipulating Standard Maclaurin Expansions • Cross-Lesson: 1.01 Proof by Mathematical Induction

In the expansion of $(1+x)^7$, find the ratio of the coefficient of $x^4$ to the coefficient of $x^3$.

**a.** 1 ✓

**b.** 2

**c.** 3

**d.** 6

Working through this complex-numbers step: $\frac{\binom{7}{4}}{\binom{7}{3}} = \frac{7-3}{3+1} = 1$

**43** [Challenging] • Target: The Ratio Test for Convergence • Cross-Lesson: 1.02 Complex Numbers: Cartesian Form & Algebra

In the expansion of $(1+x)^8$, find the ratio of the coefficient of $x^3$ to the coefficient of $x^2$.

**a.** 3

**b.** 4

**c.** 1

**d.** 2 ✓

Working through this complex-numbers step: $\frac{\binom{8}{3}}{\binom{8}{2}} = \frac{8-2}{2+1} = 2$

**44** [Challenging] • Target: Radius and Interval of Convergence • Cross-Lesson: 1.03 Polar Form, De Moivre's Theorem & Roots of Unity

Reduction formula $I_n = \int_0^{\pi/2} \sin^n x\,dx$: $I_n = \frac{n-1}{n} I_{n-2}$. Find $I_4$:

**a.** 3π/8

**b.** 3π/16 ✓

**c.** π/4

**d.** π/2

I■=(3/4)I■=(3/4)(1/2)I■=(3/4)(1/2)(π/2)=3π/16.

**45** [Challenging] • Target: The Maclaurin Series Formula • Cross-Lesson: 1.04 Partial Fractions

Let $z_1$ and $z_2$ be complex numbers such that $z_1 = 2 + i$ and $z_2 = 3 - 2i$. ; Consider the complex number $w = \frac{z_1^2}{z_2}$. ; If $w = a + bi$, where $a$ and $b$ are real numbers, find the value of $a^2 + b^2$.

**a.** $\frac{25}{13}$ ✓

**b.** $\frac{5}{13}$

**c.** $\frac{169}{25}$

**d.** $\frac{13}{5}$

First, we calculate $z_1^2$: $z_1^2 = (2+i)^2 = 4 + 4i + i^2 = 4 + 4i - 1 = 3 + 4i$. ; Next, we calculate $w = \frac{z_1^2}{z_2} = \frac{3+4i}{3-2i}$. ; To simplify, we multiply the numerator and denominator by the conjugate of the denominator: $w = \frac{(3+4i)(3+2i)}{(3-2i)(3+2i)} = \frac{9+6i+12i+8i^2}{9-4i^2} = \frac{9+18i-8}{9+4} = \frac{1+18i}{13} = \frac{1}{13} + \frac{18}{13}i$. ; So, $a = \frac{1}{13}$ and $b = \frac{18}{13}$. ; We need to find $a^2 + b^2$.

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